User Guide

Reference guide for SpaceCDF tool features

SpaceCDF -- AI-Assisted Concurrent Design Facility
University of Ottawa - SEDTI
Generated: 2026-05-10

DRAFT

SpaceCDF User Guide

Overview

SpaceCDF is an AI-assisted Concurrent Design Facility for space mission design. It guides you from problem definition through to a complete, verified CubeSat design with ECSS-compliant documentation.

Getting Started

Installation

Prerequisites: Python 3.11+, Node.js 18+

```
git clone https://github.com/FNewland/SpaceCDF.git
cd SpaceCDF

# Backend
pip install -e packages/spacecdf-common
pip install -e packages/spacecdf-agents
pip install -e packages/spacecdf-kb
pip install -e packages/spacecdf-server
uvicorn spacecdf_server.app:app --reload --port 8000

# Frontend (separate terminal)
cd frontend && npm install && npm run dev
```

Open http://localhost:5173 in your browser.

Connecting to a Remote Instance

Edit frontend/vite.config.ts and change the proxy target:

```
proxy: { '/api': 'http://REMOTE_HOST:8000' }
```

Workflow

Step 1: Mission Need

1. Click Step 1 (Need) in the left stepper
2. Enter your problem statement (WHAT, not HOW)
3. Add stakeholders with roles and needs
4. Define objectives with measurable success criteria
5. The tool auto-detects mission type from objective keywords

Step 2: Concept Exploration

1. Click Step 2 (Concept)
2. Enter trade parameters: GSD, revisit, coverage, latency, budget
3. Click "Run Analysis" to evaluate space vs non-space alternatives
4. Review scored results ? the tool indicates whether space is justified

Step 3: Requirements

1. Click Step 3 (Requirements)
2. Orbit Advisor: Enter targets, click "Compute" ? scored orbit candidates. Click "Use" to apply.
3. Class Advisor: Enter known parameters (all optional). Click "Compute" ? spacecraft class recommendation.
4. Set orbit, payload, and mission parameters in the form
5. Click "Run Design (solo)" ? 20 agents converge the design

Step 4: Design Phase

After running the design, the tool switches to the tabbed design workspace organised into 5 groups:

Design: Dashboard, ConOps, Functions, Requirements, Interfaces

Analysis: Link Budget, Trade Studies, Optimizer, Cost

Verify: Compliance, V&V Matrix, Gate Review

Team: Positions, Q&A

Data: Exports, Parametric, Changes, Help

Key Features

Dashboard

The main design view showing:

- KPI cards: Mass margin, power margin, link margin, cost, sustainability, reliability, conflicts
- Mass waterfall and power profile charts

- Budget Breakdown: Per-subsystem mass/power with editable allocations
- Engineering budgets: Pointing (RSS), Data (pipeline flow), Timing (orbit timeline)
- Spectrum Selector: Choose license type and frequency band
- Launch Selector: Pick a provider with live mass allocation update
- ECSS Margin Enforcement: Per-domain margin vs policy check

Equipment Browser

Click "Equipment" in the header bar:

- 18 categories grouped by domain (Power, AOCS, Comms, Propulsion, Structure, Data, Thermal, Integration)
- Need annotations: Blue dot = required, circle = optional, dimmed = not needed
- Multiple selection: Select multiple items per category (e.g., 4 reaction wheels)
- RF compatibility: Warning when selecting mismatched transponder/antenna bands
- Live budget bar: Running mass/power/cost totals
- Compare mode: Side-by-side comparison of up to 3 components

Trade Studies

Two modes:

- Parametric Sweep: Vary one parameter, see effect on key metrics (chart)
- Tabular Trade: Define criteria with weights, add options, score quantitatively or qualitatively (low/medium/high), see ranked results

Link Budget Tool

Full interactive cascade: TX power ? antenna gain ? losses ? FSPL ? atmospheric/pointing/polarisation ? RX gain ? G/T ? C/N? ? Eb/N? ? margin. Each term editable, margin color-coded.

Optimizer

- Single objective: Minimise mass, cost, or maximise link margin
- Pareto (NSGA-II): Multi-objective with up to 10 objectives
- Mission-type aware: Variables auto-filtered based on mission (no propulsion vars without propulsion)
- Sensitivity (Morris): Ranks which parameters most influence the objective

Exports

Generates 18+ document types:

- ECSS: MRD, TS, IRD, SEMP, RMP, ConOps, VP, Test Plan
- Regulatory: ITU API, IARU coordination, RSSSA, export control, COPUOS, EOL report
- Design data: Parametric model data, duty cycles, consistency check, margin enforcement

Cross-Tool Reactivity

- Design State Bar: Shows when design is outdated after any parameter change
- Auto-Reconverge: Optional toggle ? re-runs design automatically after each edit
- Conflict Review: Modal appears when critical conflicts detected; requires resolution
- Impact Preview: Shows which agents and budgets a change will affect before accepting
- Change Audit: Full history with undo capability

Editing Parametric Data

The tool's sizing models use CubeSat-calibrated data. To view and edit:

1. Go to Parametric tab
2. Four sub-tabs: Mass Fractions, Cost Fractions, Power Duty Cycles, SA Power
3. Values are per spacecraft class (nano, micro, small, medium, large)
4. Sources cited alongside each table

To edit the source data directly: packages/spacecdf-common/src/spacecdf_common/physics/heritage_mass.py

---

Updating Equipment Database

Component data is in YAML files: packages/spacecdf-kb/src/spacecdf_kb/data/components/

To add a component, add an entry to the appropriate YAML file:

```
- id: unique-id
  name: "Component Name"
  manufacturer: "Vendor"
  mass_kg: 0.15
  power_w: 8.0
  cost_keur: 30
  trl: 8
  frequency_band: S  # for RF components
```

The API also supports import: POST /api/lifecycle/equipment/import

---

Concurrent Design Sessions

1. Click "Start Session" in the header
2. Select your position(s) and enter a display name
3. Share the URL with team members ? they join the same session

4. Edits propagate in real-time via WebSocket
5. Each position can only edit their owned parameters
6. Convergence runs automatically after each edit

Keyboard Shortcuts

Key	Action
Ctrl+Enter	Run Design
Escape	Close modal

Troubleshooting

Issue	Solution
"Socket not connected"	Start or join a session first for equipment selection
Design shows no results	Click "Run Design" in Step 3
Requirements tab empty	Need a study ID ? run design at least once
Optimizer won't start	Need a session (or use 'solo' mode)

New Features (UP6)

Progressive Level Unlocking

The tool now follows the System-V model with 5 progressive levels:

- Level 0: Help only (define mission need first)
- Level 1: Mission Architecture (after need defined) -- ConOps, Functions, Requirements
- Level 2: System Architecture (after design run) -- Architecture options, Interfaces, Budgets, Trade Studies, Project Management
- Level 3: Subsystem Design (after architecture selected) -- Link Budget, Optimizer, Cost, Equipment Browser
- Level 4: Verification (after subsystem design) -- Compliance, V&V Matrix, Gate Review

The level indicator bar shows your current progress and what to do next.

Interactive Mission Architecture Editor

In the ConOps tab, a drag-and-drop diagram editor lets you build your mission architecture:

- 6 node types: Satellite, Ground Station, Processing, User, Sensor, GNSS/External
- Drag nodes to position, connect with labelled lines
- Architecture drives what systems need to be defined at the next level

System Architecture Selection

In the Architecture tab:

- Top half: Select architecture options for each of 8 subsystems (EPS, AOCS, TTC, Thermal, Structure, Propulsion, OBC, Ground)
- Bottom half: Auto-generated block diagram showing subsystems and their connections
- Each selection derives requirements (tagged performance/interface/budget/functional)

Engineering Budgets (Unified View)

New "Eng. Budgets" tab shows all 8 budgets in one place:

- Mass, Power, Link, Pointing, Delta-V, Volume, Data, Cost
- Configurable margins by design phase (Phase A: 20%, B: 15%, C: 10%)
- Per-subsystem breakdown with margin source (COTS vs new design)
- Roll-up from equipment selections

Constraint Propagation Engine

187 design point interconnections detected automatically:

- When any parameter changes, the engine identifies ALL affected budgets
- Shows resolution options for violations with cross-budget trade-off analysis
- Detects circular dependencies (trade-off loops requiring team decision)

Project Management

New "Project Mgmt" tab with:

- 5x5 Risk Matrix: Interactive, color-coded by LxC score
- Schedule: 10 milestones from MCR through commissioning
- WBS: 9 work packages with effort hours and status tracking
- Project Manager position added

Word Document Export

ECSS documents can now be exported as editable .docx files:

- Click "Word" button next to MRD, ConOps, or VP in the Exports tab
- Documents populated from live design state
- Editable in Microsoft Word or LibreOffice

Session Persistence

Your design state now persists across page refreshes:

- All data saved to browser localStorage automatically
- No need to "save" -- it's always saved
- To clear: browser DevTools -> Application -> Local Storage -> delete 'spacecdf-design-state'